Linear Response Theory and the Lindhard Polarizability of the Three-Dimensional Electron Gas

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Abstract

We present a self-contained derivation of the dynamical and static Lindhard polarizability $\chi_0(\mathbf{q}, \omega)$ of a three-dimensional homogeneous free electron gas, starting from the time-dependent Schrödinger equation. In Section 1, we formulate the interaction picture on a general Hilbert space, derive the first-order Kubo linear response commutator for a time-dependent perturbation, and evaluate the frequency-space response under adiabatic switching to obtain the two-pole Lehmann representation. In Section 2, we specialize the abstract framework to the second-quantized many-body jellium model and reduce the density-density response to a single-particle integral over the Fermi sea. In Section 3, we evaluate the continuous momentum integral for a parabolic band $\varepsilon(\mathbf{k}) = \|\mathbf{k}\|^2/(2m)$ and derive the exact radial primitive in the static limit, exhibiting the logarithmic singularity at $q = 2k_F$ responsible for Friedel oscillations and the Kohn anomaly.

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1 The Interaction Picture and Kubo Linear Response

Consider a quantum system whose state vectors reside in a complex Hilbert space $\mathcal{H}$ equipped with the inner product $\langle \phi, \psi \rangle$, taken to be conjugate-linear in the first argument and linear in the second argument. We work in natural units where $\hbar = 1$.

1.1 Free Evolution and the Interaction Picture

Let the total Hamiltonian of the system be decomposed into a time-independent unperturbed Hamiltonian H_0 and a time-dependent perturbation $V(t)$:

$$H(t) = H_0 + V(t). \tag{1.1}$$

In the Schrödinger picture, the state trajectory $\Psi_S(t)$ evolves according to the time-dependent Schrödinger equation:

$$\frac{\mathrm{d}}{\mathrm{d}t}\Psi_S(t) = -i\big(H_0\Psi_S(t) + V(t)\Psi_S(t)\big). \tag{1.2}$$

To isolate the dynamics generated specifically by the perturbation $V(t)$, we introduce the unperturbed evolution operator $U_0(t)$ defined via the operator exponential:

$$U_0(t) := \exp(-iH_0t) = \sum_{n=0}^{\infty} \frac{(-itH_0)^n}{n!}. \tag{1.3}$$

Because $t(-iH_0)$ and $(-t)(-iH_0)$ commute for any t, the exponential addition law implies the group inversion identity:

$$U_0(t)U_0(-t) = \exp\big(-iH_0t + iH_0t\big) = \exp(0) = \mathbf{1}. \tag{1.4}$$

Differentiating the power series (1.3) with respect to t yields the forward and backward equations of motion for the free propagator:

$$\frac{\mathrm{d}}{\mathrm{d}t}U_0(t) = -i\,U_0(t)H_0, \qquad \frac{\mathrm{d}}{\mathrm{d}t}U_0(-t) = +i\,U_0(-t)H_0, \tag{1.5}$$

where the second identity follows from the chain rule under $t \mapsto -t$.

We now pass to the *interaction picture* by transforming state vectors and operators via the backward propagator $U_0(-t)$:

$$\Psi_I(t) := U_0(-t)\Psi_S(t), \tag{1.6}$$
$$V_I(t) := U_0(-t)\,V(t)\,U_0(t), \tag{1.7}$$
$$A_I(t) := U_0(-t)\,A\,U_0(t), \tag{1.8}$$

where A is any time-independent Schrödinger-picture observable. Differentiating $\Psi_I(t)$ with respect to t via the Leibniz product rule and substituting Eqs. (1.2) and (1.5) cancels the free drift H_0:

$$\begin{aligned}\frac{\mathrm{d}}{\mathrm{d}t}\Psi_I(t) &= \left(\frac{\mathrm{d}}{\mathrm{d}t}U_0(-t)\right)\Psi_S(t) + U_0(-t)\left(\frac{\mathrm{d}}{\mathrm{d}t}\Psi_S(t)\right)\\ &= +iU_0(-t)H_0\Psi_S(t) + U_0(-t)\big(-iH_0\Psi_S(t) - iV(t)\Psi_S(t)\big)\\ &= -iU_0(-t)V(t)\Psi_S(t).\end{aligned} \tag{1.9}$$

Inserting the identity $\mathbf{1} = U_0(t)U_0(-t)$ from Eq. (1.4) between $V(t)$ and $\Psi_S(t)$ yields the interaction-picture equation of motion:

$$\frac{\mathrm{d}}{\mathrm{d}t}\Psi_I(t) = -iU_0(-t)V(t)U_0(t)U_0(-t)\Psi_S(t) = -i\,V_I(t)\Psi_I(t). \tag{1.10}$$

1.2 First-Order Perturbation Theory and the Kubo Commutator

To extract the linear response to the perturbation, we rescale $V(t) \mapsto gV(t)$ by a dimensionless coupling parameter g and denote the corresponding state at time t by $\Phi(g)$, with unperturbed initial state $\Phi(0) = \psi_0$ and first-order variation $\delta\Phi = \frac{\mathrm{d}\Phi}{\mathrm{d}g}\Big|_{g=0}$. Differentiating the interaction-picture generator $-i\big(gV_I(t)\big)\Phi(g)$ with respect to g at $g = 0$ by the product rule isolates the first-order source term:

$$\frac{\mathrm{d}}{\mathrm{d}g}\Big(- igV_I(t)\Phi(g)\Big)\Big|_{g=0} = -i\,(V_I(t)\Phi(0) + 0 \cdot V_I(t)\delta\Phi) = -i\,V_I(t)\psi_0. \tag{1.11}$$

Thus, to first order in the perturbation, the state trajectory satisfies $\frac{\mathrm{d}}{\mathrm{d}s}\delta\Psi_I(s) = -iV_I(s)\psi_0$, so each past instant $s \le t$ contributes an infinitesimal state variation:

$$\delta\psi(s) = -i\,V_I(s)\psi_0. \tag{1.12}$$

Consider the expectation value $\langle \Psi_I(t), A_I(t)\Psi_I(t)\rangle$ of an observable A at time t. A first-order variation $\delta\psi = -iV_I(s)\psi_0$ of the state produces the linear shift:

$$\delta\langle A(t)\rangle_s = \langle \delta\psi, A_I(t)\psi_0\rangle + \langle \psi_0, A_I(t)\delta\psi\rangle. \tag{1.13}$$

Assuming the perturbation $V_I(s)$ is self-adjoint ($\langle V_I(s)x, y\rangle = \langle x, V_I(s)y\rangle$ for all $x, y \in \mathcal{H}$), we use the conjugate-linearity of the inner product in the first slot (which pulls out $-i$ as $+i$) and linearity in the second slot:

$$\begin{aligned}\langle -iV_I(s)\psi_0, A_I(t)\psi_0\rangle + \langle \psi_0, A_I(t)(-iV_I(s)\psi_0)\rangle &= +i\langle V_I(s)\psi_0, A_I(t)\psi_0\rangle - i\langle \psi_0, A_I(t)V_I(s)\psi_0\rangle \\ &= +i\langle \psi_0, V_I(s)A_I(t)\psi_0\rangle - i\langle \psi_0, A_I(t)V_I(s)\psi_0\rangle \\ &= -i\,\big\langle \psi_0, [A_I(t), V_I(s)]\psi_0\big\rangle, \end{aligned} \tag{1.14}$$

where $[X, Y] := XY - YX$. Integrating this commutator over the history $s \in (-\infty, t]$ yields the time-domain Kubo formula:

$$\delta\langle A(t)\rangle = -i\int_{-\infty}^{t} \mathrm{d}s\ \big\langle \psi_0, [A_I(t), V_I(s)]\psi_0\big\rangle. \tag{1.15}$$

1.3 Adiabatic Switching and the Two-Pole Lehmann Representation

Suppose the perturbation is switched on adiabatically from the distant past $s \to -\infty$ with complex frequency $z = \omega + i\eta$ ($\eta > 0$). For a transition between the ground state ψ_0 (with energy E_0) and an excited eigenstate component ϕ (with energy E_n and Bohr transition frequency $\Delta E = E_n - E_0$), the interaction-picture source oscillates in time as:

$$\frac{\mathrm{d}}{\mathrm{d}s}\delta\Psi_\phi(s) = \big(- i\,e^{-i(z-\Delta E)s}\big)\phi. \tag{1.16}$$

Provided $z - \Delta E \neq 0$ (which is guaranteed for all real ΔE whenever $\eta > 0$), differentiation with respect to s confirms the exact integrated state trajectory:

$$\frac{\mathrm{d}}{\mathrm{d}s}\left(\frac{e^{-i(z-\Delta E)s}}{z-\Delta E}\,\phi\right) = \frac{-i(z-\Delta E)e^{-i(z-\Delta E)s}}{z-\Delta E}\,\phi = \big(- i\,e^{-i(z-\Delta E)s}\big)\phi. \tag{1.17}$$

Because $\eta > 0$, the exponential vanishes at the lower limit $s \to -\infty$. Factoring out the overall harmonic time dependence e^{-izt} (or evaluating the integrated amplitude at $t = 0$), the resonant ket shift and anti-resonant bra shift take the form:

$$\delta\psi_{\mathrm{ket}} = \frac{1}{z-\Delta E}\,\phi, \qquad \delta\psi_{\mathrm{bra}} = -\frac{1}{\bar{z}+\overline{\Delta E}}\,\phi. \tag{1.18}$$

Substituting these amplitudes into the expectation-value variation of A and using the conjugate-linearity of the first inner-product slot ($\overline{-(\bar{z}+\overline{\Delta E})^{-1}} = -(z+\Delta E)^{-1}$) yields the two-pole Lehmann susceptibility for the transition mode:

$$\langle \delta\psi_{\mathrm{bra}}, A\psi_0\rangle + \langle \psi_0, A\delta\psi_{\mathrm{ket}}\rangle = \frac{\langle \psi_0, A\phi\rangle}{z-\Delta E} - \frac{\langle \phi, A\psi_0\rangle}{z+\Delta E}. \tag{1.19}$$

2 Application to the Homogeneous Free Electron Gas

We now specialize the general linear response formalism of Section 1 to the density response of a three-dimensional non-interacting electron gas (jellium) of particle mass m and Fermi wavenumber k_F.

2.1 Second-Quantized Hamiltonian and Density Perturbation

In second quantization, the many-body fermionic Fock space is generated by creation and annihilation operators $\hat{c}^\dagger_{\mathbf{k}}, \hat{c}_{\mathbf{k}}$ indexed by continuous momentum $\mathbf{k} \in \mathbb{R}^3$ (suppressing spin degeneracy, which contributes an overall factor of 2), obeying the canonical anticommutation relations:

$$\left\{\hat{c}_{\mathbf{k}}, \hat{c}^\dagger_{\mathbf{k}'}\right\} = (2\pi)^3\delta^{(3)}(\mathbf{k}-\mathbf{k}'), \qquad \{\hat{c}_{\mathbf{k}}, \hat{c}_{\mathbf{k}'}\} = \left\{\hat{c}^\dagger_{\mathbf{k}}, \hat{c}^\dagger_{\mathbf{k}'}\right\} = 0. \tag{2.1}$$

The unperturbed Hamiltonian H_0 describes free fermions with parabolic dispersion $\varepsilon(\mathbf{k})$:

$$H_0 = \int \frac{\mathrm{d}^3\mathbf{k}}{(2\pi)^3}\, \varepsilon(\mathbf{k})\, \hat{c}^\dagger_{\mathbf{k}}\hat{c}_{\mathbf{k}}, \qquad \varepsilon(\mathbf{k}) = \frac{\|\mathbf{k}\|^2}{2m}. \tag{2.2}$$

The unperturbed ground state $\psi_0 = |\mathrm{FS}\rangle$ is the filled Fermi sphere, characterized by the occupation number expectation value:

$$\langle \psi_0, \hat{c}^\dagger_{\mathbf{k}}\hat{c}_{\mathbf{k}'}\psi_0\rangle = (2\pi)^3\delta^{(3)}(\mathbf{k}-\mathbf{k}')\, n_F(\mathbf{k}), \qquad n_F(\mathbf{k}) = \Theta(k_F - \|\mathbf{k}\|). \tag{2.3}$$

We subject the electron gas to an external scalar potential with Fourier amplitude $\phi_{\mathrm{ext}}(\mathbf{q}, z)$ at wavevector $\mathbf{q}$ and complex frequency $z = \omega + i\eta$, which couples to the Fourier component of the electron density operator:

$$A = \hat{\rho}(\mathbf{q}) = \int \frac{\mathrm{d}^3\mathbf{k}}{(2\pi)^3}\, \hat{c}^\dagger_{\mathbf{k}-\mathbf{q}}\hat{c}_{\mathbf{k}}, \qquad V(t) = e^{-izt}\, \phi_{\mathrm{ext}}(\mathbf{q}, z)\, \hat{\rho}(-\mathbf{q}). \tag{2.4}$$

2.2 Particle-Hole Excitations and Pauli Blocking Cancellation

Acting with $\hat{\rho}(-\mathbf{q}) = \int \frac{\mathrm{d}^3\mathbf{k}}{(2\pi)^3}\hat{c}^\dagger_{\mathbf{k}+\mathbf{q}}\hat{c}_{\mathbf{k}}$ on the ground state ψ_0 annihilates a fermion at momentum $\mathbf{k}$ inside the Fermi sea and creates a fermion at momentum $\mathbf{k}+\mathbf{q}$ outside the Fermi sea. The corresponding particle-hole excitation energy is:

$$\Delta E_{\mathbf{k},\mathbf{q}} = \varepsilon(\mathbf{k}+\mathbf{q}) - \varepsilon(\mathbf{k}). \tag{2.5}$$

By the Pauli exclusion principle, the resonant transition $\mathbf{k} \to \mathbf{k}+\mathbf{q}$ carries weight $n_F(\mathbf{k})\big(1 - n_F(\mathbf{k}+\mathbf{q})\big)$, while the anti-resonant de-excitation term in Eq. (1.19) carries weight $n_F(\mathbf{k})\big(1 - n_F(\mathbf{k}-\mathbf{q})\big)$ with energy difference $\varepsilon(\mathbf{k}) - \varepsilon(\mathbf{k}-\mathbf{q})$. Integrating Eq. (1.19) over all momenta $\mathbf{k}$ yields the dynamical density response function (polarizability):

$$\chi_0(\mathbf{q}, z) = \int \frac{\mathrm{d}^3\mathbf{k}}{(2\pi)^3}\left[\frac{n_F(\mathbf{k})\big(1-n_F(\mathbf{k}+\mathbf{q})\big)}{z - \big(\varepsilon(\mathbf{k}+\mathbf{q}) - \varepsilon(\mathbf{k})\big)} - \frac{n_F(\mathbf{k})\big(1-n_F(\mathbf{k}-\mathbf{q})\big)}{z - \big(\varepsilon(\mathbf{k}) - \varepsilon(\mathbf{k}-\mathbf{q})\big)}\right]. \tag{2.6}$$

Notice that when we shift the integration variable $\mathbf{k} \mapsto \mathbf{k}+\mathbf{q}$ in the second term, the quadratic occupation product $-n_F(\mathbf{k})n_F(\mathbf{k}+\mathbf{q})$ appears with identical denominators in both terms and cancels identically. Consequently, we may drop the quadratic Pauli-blocking product from the outset and write the two-pole susceptibility directly in terms of linear occupations $n_F(\mathbf{k})$.

(Equivalently, evaluating the equal-momentum commutator $\left[\hat{c}^\dagger_{\mathbf{k}}\hat{c}_{\mathbf{k}+\mathbf{q}}, \hat{c}^\dagger_{\mathbf{k}+\mathbf{q}}\hat{c}_{\mathbf{k}}\right] = \hat{c}^\dagger_{\mathbf{k}}\hat{c}_{\mathbf{k}} - \hat{c}^\dagger_{\mathbf{k}+\mathbf{q}}\hat{c}_{\mathbf{k}+\mathbf{q}}$ *directly inside the Kubo formula (1.14) produces the linear difference* $n_F(\mathbf{k}) - n_F(\mathbf{k}+\mathbf{q})$ *without intermediate quadratic terms.)*

3 Evaluating the Three-Dimensional Lindhard Integral

3.1 Parabolic Band Kinematics and Momentum-Space Translation

For the quadratic dispersion $\varepsilon(\mathbf{k}) = \|\mathbf{k}\|^2/(2m)$ on a real inner product space, expanding the squared norm $\|\mathbf{k}+\mathbf{q}\|^2 = \|\mathbf{k}\|^2 + 2\langle \mathbf{k}, \mathbf{q}\rangle + \|\mathbf{q}\|^2$ cancels the $\|\mathbf{k}\|^2/(2m)$ kinetic energy and gives the particle-hole excitation energy:

$$\varepsilon(\mathbf{k}+\mathbf{q}) - \varepsilon(\mathbf{k}) = \frac{\|\mathbf{k}+\mathbf{q}\|^2 - \|\mathbf{k}\|^2}{2m} = \frac{2\langle \mathbf{k}, \mathbf{q}\rangle + \|\mathbf{q}\|^2}{2m}. \tag{3.1}$$

Starting from the two-pole integral representation (suppressing the overall phase-space factor $(2\pi)^{-3}$ for clarity):

$$\chi_{\text{two-pole}}(\mathbf{q}, z) = \int \mathrm{d}\mathbf{k}\, \frac{n_F(\mathbf{k})}{z - \big(\varepsilon(\mathbf{k}+\mathbf{q}) - \varepsilon(\mathbf{k})\big)} - \int \mathrm{d}\mathbf{k}\, \frac{n_F(\mathbf{k})}{z - \big(\varepsilon(\mathbf{k}) - \varepsilon(\mathbf{k}-\mathbf{q})\big)}, \tag{3.2}$$

where both integrals are separately integrable for $\operatorname{Im} z = \eta > 0$ since $n_F(\mathbf{k})$ has compact support inside the Fermi sphere $\|\mathbf{k}\| \le k_F$. By translation invariance of the Lebesgue measure ($\mathrm{d}(\mathbf{k}+\mathbf{q}) = \mathrm{d}\mathbf{k}$), we shift the integration variable $\mathbf{k} \mapsto \mathbf{k}+\mathbf{q}$ in the second integral:

$$\int \mathrm{d}\mathbf{k}\, \frac{n_F(\mathbf{k})}{z - \big(\varepsilon(\mathbf{k}) - \varepsilon(\mathbf{k}-\mathbf{q})\big)} = \int \mathrm{d}\mathbf{k}\, \frac{n_F(\mathbf{k}+\mathbf{q})}{z - \big(\varepsilon(\mathbf{k}+\mathbf{q}) - \varepsilon(\mathbf{k})\big)}. \tag{3.3}$$

Combining the two integrals over their common denominator and substituting the kinematic identity (3.1) yields the canonical Lindhard polarizability integral:

$$\begin{aligned} \chi_0(\mathbf{q}, z) &= \int \mathrm{d}\mathbf{k}\, \frac{n_F(\mathbf{k}) - n_F(\mathbf{k}+\mathbf{q})}{z - \big(\varepsilon(\mathbf{k}+\mathbf{q}) - \varepsilon(\mathbf{k})\big)} \\ &= \int \mathrm{d}\mathbf{k}\, \frac{n_F(\mathbf{k}) - n_F(\mathbf{k}+\mathbf{q})}{z - \dfrac{2\langle \mathbf{k}, \mathbf{q}\rangle + \|\mathbf{q}\|^2}{2m}}. \end{aligned} \tag{3.4}$$

3.2 Spherical Coordinates and Polar Angle Integration in the Static Limit

We now evaluate $\chi_0(\mathbf{q}, z)$ in three dimensions in the static limit $z \to 0$. Using the symmetry $\mathbf{k} \mapsto -\mathbf{k}$ of the Fermi sphere $n_F(\mathbf{k}) = \Theta(k_F - \|\mathbf{k}\|)$, the two terms in Eq. (3.2) give equal contributions at $z = 0$:

$$\chi_0(\mathbf{q}, 0) = -4m \int_{\|\mathbf{k}\| \le k_F} \frac{\mathrm{d}^3\mathbf{k}}{(2\pi)^3}\, \frac{1}{2\langle \mathbf{k}, \mathbf{q}\rangle + q^2}, \tag{3.5}$$

where $q = \|\mathbf{q}\| > 0$. Introducing spherical coordinates (k, θ, ϕ) with the polar axis parallel to $\mathbf{q}$, we set $u = \cos\theta \in [-1, 1]$ so that $\langle \mathbf{k}, \mathbf{q}\rangle = kqu$ and $\mathrm{d}^3\mathbf{k} = 2\pi k^2\, \mathrm{d}k\, \mathrm{d}u$.

Performing the elementary integration over $u \in [-1, 1]$ first gives:

$$\int_{-1}^{1} \frac{\mathrm{d}u}{2kqu + q^2} = \left[\frac{1}{2kq} \ln\left|2kqu + q^2\right|\right]_{u=-1}^{u=1} = \frac{1}{2kq}\Big(\ln|2k+q| - \ln|2k-q|\Big). \tag{3.6}$$

Multiplying by the radial Jacobian $k^2\,\mathrm{d}k$ cancels the $1/k$ prefactor:

$$\chi_0(q,0) = -\frac{m}{2\pi^2 q}\int_0^{k_F} \mathrm{d}k\; k\Big(\ln|2k+q| - \ln|2k-q|\Big). \tag{3.7}$$

3.3 Exact Radial Primitive and the $q = 2k_F$ Singularity

To evaluate the remaining radial integral in Eq. (3.7), we introduce the radial primitive function:

$$F(q,k) := \frac{1}{2}\left(k^2 - \frac{q^2}{4}\right)\Big(\ln|2k+q| - \ln|2k-q|\Big) + \frac{1}{2}qk. \tag{3.8}$$

We verify that $\frac{\mathrm{d}}{\mathrm{d}k}F(q,k) = k\big(\ln|2k+q| - \ln|2k-q|\big)$ for all $2k+q \neq 0$ and $2k-q \neq 0$ by differentiating each component:

1. The derivative of the quadratic polynomial factor is:

$$\frac{\mathrm{d}}{\mathrm{d}k}\left[\frac{1}{2}\left(k^2 - \frac{q^2}{4}\right)\right] = k. \tag{3.9}$$

2. The derivative of the logarithmic difference is:

$$\frac{\mathrm{d}}{\mathrm{d}k}\Big[\ln|2k+q| - \ln|2k-q|\Big] = \frac{2}{2k+q} - \frac{2}{2k-q} = \frac{-4q}{(2k+q)(2k-q)}. \tag{3.10}$$

3. The derivative of the linear counterterm is:

$$\frac{\mathrm{d}}{\mathrm{d}k}\left[\frac{1}{2}qk\right] = \frac{1}{2}q. \tag{3.11}$$

Applying the product rule to $F(q,k)$ and factoring $k^2 - \frac{q^2}{4} = \frac{(2k+q)(2k-q)}{4}$ gives:

$$\begin{aligned}
\frac{\mathrm{d}}{\mathrm{d}k}F(q,k) &= k\Big(\ln|2k+q| - \ln|2k-q|\Big) + \frac{1}{2}\left(\frac{(2k+q)(2k-q)}{4}\right)\left(\frac{-4q}{(2k+q)(2k-q)}\right) + \frac{1}{2}q \\
&= k\Big(\ln|2k+q| - \ln|2k-q|\Big) + \left(-\frac{4q}{8}\right)\frac{(2k+q)(2k-q)}{(2k+q)(2k-q)} + \frac{1}{2}q \\
&= k\Big(\ln|2k+q| - \ln|2k-q|\Big) - \frac{1}{2}q + \frac{1}{2}q \\
&= k\ln\left|\frac{2k+q}{2k-q}\right|.
\end{aligned} \tag{3.12}$$

At the lower integration limit $k = 0$, the primitive vanishes identically because $\ln|q| = \ln|-q|$:

$$F(q,0) = -\frac{q^2}{8}\Big(\ln|q| - \ln|-q|\Big) + 0 = 0. \tag{3.13}$$

Evaluating $F(q,k_F) - F(q,0)$ inside Eq. (3.7) (and restoring a factor of 2 for electron spin) yields the static 3D Lindhard polarizability:

$$\chi_0(q,0) = -\frac{mk_F}{2\pi^2}\left[1 + \frac{k_F}{q}\left(1 - \frac{q^2}{4k_F^2}\right)\ln\left|\frac{2k_F+q}{2k_F-q}\right|\right]. \tag{3.14}$$

At the spanning wavevector $q = 2k_F$ connecting opposite antipodal points on the Fermi sphere, the argument $2k_F - q$ of the logarithm vanishes and the slope $\frac{\mathrm{d}\chi_0(q,0)}{\mathrm{d}q}$ diverges logarithmically, giving rise to the Kohn anomaly in lattice phonon spectra and oscillatory Friedel screening $\delta\rho(r) \propto \cos(2k_F r)/r^3$ at large distances.